



Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

COS20028 Big Data Architecture and Application Lab Task 1 - Semester 2, 2024

Assessment Title: Write a MapReduce Program for Finding the Shifted Word Counts

Assessment Outcome: Pass/Fail (for practice)

Assessable Item:

- One (1) piece of a report containing the complete codes of the driver, mapper, and reducer with the screenshot of submitted commands and execution results.

Purpose of Assignment

This assignment evaluates whether a student understands how the MapReduce works and knowing where to make corresponding changes for implementing a customised function.

This is an individual assignment and is mandatory for passing this unit.

Introduction

The source data is built in the virtual machine image under the path of `/home/training/training_materials/developer/data`. Our goal is to find the word count **shifted minus twenty (-20)** from the “`Shakespeare.tar.gz`” file using MapReduce.

Assignment Goal

This assignment aims at training students to get familiar with the major components in MapReduce, recognising the functionalities of different components, and identifying the right place for implementing functions to accommodate the requirements.

Assignment Task

Write a MapReduce program to find the shifted word count of the input source. Upper case and lower case words are treated as different items. The assigned shift value is minus twenty (-20). For example, assume the word “Alpha” appears sixteen (16) times in the input source, the shifted word count result for this word would be minus four (-4). You must use `System.out.print()` or `System.out.println()` to print your name and student ID on the screen while executing from the driver code. Please use the template to prepare your submission.

Submission Requirement

Only the submission using the given template will be evaluated by the tutors.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

Rubric: Lab Task 1

Task	Assessment Outcome
Successfully finding the answer to the given question in the template and the submitted code in the report is reasonable, complete, and fully functional.	Pass/Fail

----- End of Assignment -----